

Module 5, Read 0

Review of scatterplots and correlation and Preview of Project 5

Module 5, Read 0: Prediction

In order to use data to make a prediction we need to *model* the relationship between what we want to predict and what we have to predict from. In this course, we only see linear model so first we'll review the skill of assessing linearity.

Project: [Project 5, Looking Ahead](#) · Next: [Read 1](#)

i Note:

This is just a five-minute review, not new material. You already did scatterplots and Pearson correlation in [Module 3 Read 2](#). Module 5 builds directly on them: in Read 1 we go from *describing* a linear relationship to *fitting a line and predicting* with it.

In Module 5 you will fit a line to two numerical variables and use it to predict. That only makes sense when the relationship is roughly linear, and the tools you use to decide that are exactly the ones from Module 3:

- the scatterplot (to see the form), and
- the Pearson correlation r (to summarize the strength and direction of a *linear* form).

Reading a scatterplot: form, direction, strength

For two numerical variables, put the explanatory variable on the x -axis (what we have to predict from) and the response on the y -axis (what we want to predict), then describe three things:

- Form, linear, curved, or no clear pattern.
- Direction, positive (up to the right), negative (down to the right), or none.
- Strength, how tightly the points cluster around the trend (strong, moderate, weak).

Always note unusual points (outliers) too, since they can distort everything that follows.

Pearson correlation r

Pearson correlation r is a single, unitless number between -1 and 1 that summarizes the direction and strength of a *linear* relationship:

- The sign gives direction: $r > 0$ positive, $r < 0$ negative.
- The magnitude $|r|$ gives strength: near 1 is a tight linear trend, near 0 is little or no *linear* trend.

For sample data (x_i, y_i) , $i = 1, \dots, n$:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

The numerator is large and positive when x and y tend to be above (or below) their means together; the denominator rescales by the spread in each variable so the result always lands between -1 and 1 .

⚠ r only describes a *linear* trend

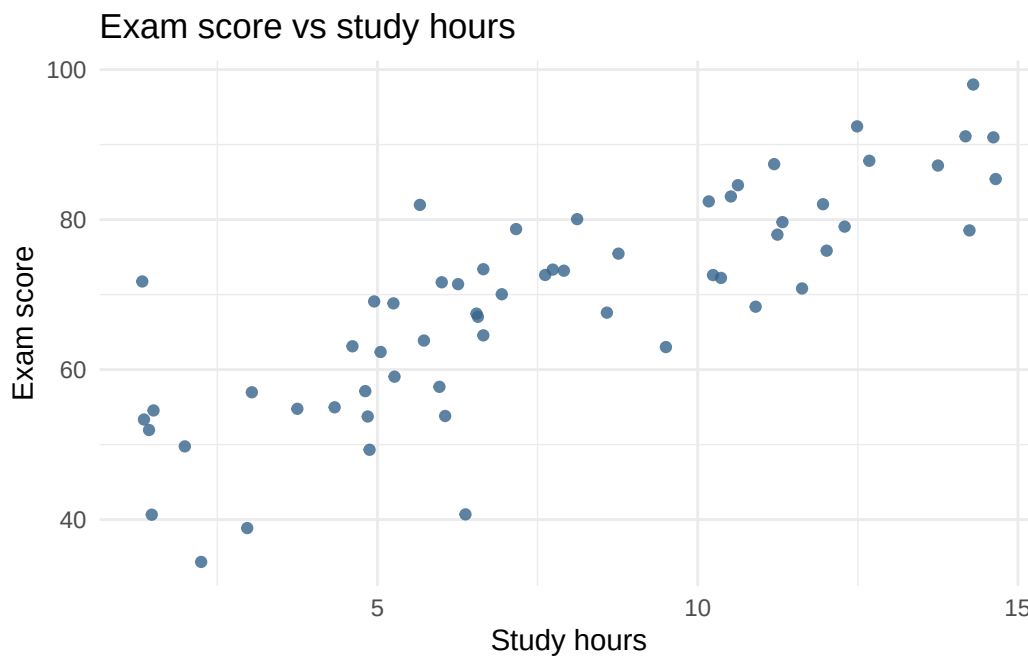
A correlation near 0 does not mean “no relationship”, it means “no *linear* relationship.” A strong curved pattern (a U-shape, say) can have $r \approx 0$. And a single outlier can pull r toward or away from ± 1 . Always look at the scatterplot before trusting r .

In R: `cor()` and a scatterplot

```
library(tidyverse)

set.seed(509)
review_dat <- tibble(
  study_hours = runif(60, 1, 15),
  exam_score = 45 + 3.4 * study_hours + rnorm(60, 0, 8)
)

ggplot(review_dat, aes(x = study_hours, y = exam_score)) +
  geom_point(alpha = 0.8, color = "steelblue4") +
  labs(title = "Exam score vs study hours",
       x = "Study hours", y = "Exam score") +
  theme_minimal()
```



```
cor(review_dat$study_hours, review_dat$exam_score)
```

```
[1] 0.817906
```

Here the form is roughly linear, the direction is positive, and r is moderately strong, exactly the situation where fitting a line (Read 1) is reasonable.

Check yourself, scatterplots and correlation

1. A scatterplot of two variables shows points falling tightly along a downward line. Describe form, direction, and strength, and give a plausible value of r .
2. A researcher reports $r = 0.05$ and concludes “these variables are unrelated.” Why might that conclusion be wrong?
3. In R, what does `cor(x, y)` return, and what range can it take?

Answers, scatterplots and correlation

1. Form: linear. Direction: negative. Strength: strong. A plausible value is $r \approx -0.9$.
 2. r only measures *linear* association. A strong curved relationship (or one masked by an outlier) can give $r \approx 0$, so the variables may be strongly related in a non-linear way. Look at the scatterplot.
 3. `cor(x, y)` returns the Pearson correlation between two numerical vectors, a unitless number between -1 and 1 .
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Preview of Project 5

Project 5, Looking Ahead is a prediction challenge: you collect your own data with the **stone jar sampler**, fit a simple linear regression with `lm()`, and report a prediction with an 95% prediction interval for the number of stones in Rosanna's cup.

What to do next

Start **Read 1**, simple linear regression, inference for the slope and correlation, prediction quality, and confidence and prediction intervals. When you finish, take **Quiz A**.